MILITARY SPECIFICATION

TUBE, LASER, PLASMA, HELIUM NEON: 11746797

WARNING EYE HAZARD

THE LASER BEAM CAN BE DANGEROUS AND CAUSE BLINDNESS IF IT ENTERS THE EYES EITHER DIRECTLY OR REFLECTED OFF OF A REFLECTED SURFACE. BEFORE FIRING THE LASER, OPERATOR SHOULD TAKE ADEQUATE PRECAUTIONS TO ASSURE MAXIMUM SAFETY.

This Specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. - This specification covers a helium neon plasma laser tube consisting of a plasma tube, a ballast resistor, and high voltage plug-in connectors, encapsulated in an austenitic corrosion-resistant steel cylindrical case (see 6.1).

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein.

SPECIFICATIONS

Military

MIL-E-75	Electron Tubes, Preparation for Delivery of
MIL-F-13926	Fire Control Material, General Specification Governing the Manufacture and Inspection of
MIL-T-23991	Training Devices, Military; General Specification for
MIL-I-45607	Inspection Equipment; Acquisition Maintenance, and Disposition of

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commander, Frankford Arsenal, ATTN: SARFA-MDM, Phila., Pa. 19137 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

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STANDARDS

MILITARY

MIL-STD-105	Sampling Procedures and Tables for Inspection By Attributes
MIL-STD-109	Quality Assurance Terms and Definitions

DRAWINGS

B11739401	Gage, Interchangeability
BI1739402	Gage, Interchangeability
F11739861	Power supply
F11746797	Tube, Laser, Plasma, Helium Neon

(Copies of specifications, standards, drawings, and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer).

3. REQUIREMENTS

3.1 Fabrication. - The tube shall remanufactured in accordance with Drawing F11746797 and related documents thereto. The tube assembly design and final configuration shall also comply with the requirements specified in 3.1.1 through 3.1.8.

3.1.1 Materials. - The materials shall be in accordance with MIL-T-23991 as applicable as well as material specifications, general specifications and figures forming a part of this specification.

3.1.2 Encapsulating compound. - The encapsulating compound shall be a stable shock absorbing material capable-of maintaining alignment between the plasma tube and the inside diameter of the cylindrical case for the specified life of the laser tube, under all specified conditions, or combinations thereof. Dielectric strength of encapsulating compound shall prevent arcing or corona between electrodes and between each electrode and the cylindrical case. Encapsulating compound shall be in accordance with impregnating, filling, potting, and encapsulating compounds of MIL-T-23991.

3.1.3 Fungus-inert materials. - Only fungus-inert materials in accordance with MIL-T-23991 shall be used.

3.1.4 Internal mirrors. - The plasma tube shall have permanent, factory-aligned, internal mirrors requiring no alignment or other maintenance.

3.1.5 Coaxial reservoir. - The plasma tube shall utilize a coaxial reservoir to reduce the space requirements and to improve ruggedness and life.

3.1.6 Cold cathode. - The plasma tube shall utilize a cold cathode.

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3.1.7 Ballast resistor. - A ballast resistor shall be provided as part of the laser thbe assembly. The value of the ballast resistor shall be selected to achieve the operating characteristics of Figure 2 when using a power supply that has 70k ± 10% ballast resistance.

3.1.8 Connectors. - Electrical connections shall be by means of high-voltage connectors encapsulated into the rear of the laser tube assembly. The connectors shall conform to Drawing 11737748.

3.2 General specifications. - The tube shall meet the following requirements of Specification MIL-F-13926:

(a) Order of precedence

(b) Dimensions and tolerances

(c) Inorganic protective surface finishes

(d) Workmanship

3.3 Reliability. - The failure rate shall not exceed 200 failures per million hours in the continuous mode (50 minutes on and 10 minutes off.)

3.4 Pre-encapsulation. - Prior to encapsulation, the plasma tube shall meet the requirements of 3.4.1 and 3.4.2. Only those tubes that have successfully passed 3.4.1, 3.4.2 shall be encapsulated and shall then meet all the other requirements of this specification.

3.4.1 Leakage. - The unencapsulated tube shall withstand four hours of immersion in boiling water (212°F) without damage and subsequently meet the performance requirements of 3.4.2.

3.4.2 Burn-in (unencapsulated). - The unencapsulated tube after successfully meeting the immersion requirement of 3.4.1 shall be operated through 24 cycles of 50 minutes on and 10 minutes off and subsequently meet the performance requirements of 3.7.1 through 3.7.5, inclusive.

3.5 Encapsulation. - After the plasma tube is encapsulated, the tube assembly shall meet the requirements of 3.7.1 through 3.7.10 inclusive at room ambient temperature prior to being subjected to 3.5.2.

3.5.1 Sealing. - The tube assembly shall show no evidence of leakage in excess of 0.2 psig when subjected to an internal pressure of 5 psig at the output end of the tube assembly for a minimum of two hours.

3.5.2 Burn-in encapsulated. - The encapsulated tube shall be operated through 72 cycles of 50 minutes on and 10 minutes off and subsequently shall meet the requirements of 3.7, 3.8 and 3.9.

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3.6 Environmental requirements. -The laser tube assembly, when subjected to the environmental tests specified herein, shall be mounted as shown in Figure 2, and shall not be damaged, degraded or rendered inoperative by the specified conditions.

3.6.1 Storage temperature. - The tube assembly shall show no evidence of damage or degradation of performance after being subjected to storage temperatures of -65°F and +155°F.

3.6.2 Operating temperature. - The tube assembly shall meet all performance requirements of this specification while exposed and thermally stabilized at temperatures of 0°F and 100°F.

3.6.3 Vibration. - The tube assembly shall be mounted as shown in Figure 2, and vibrated in two directions parallel and perpendicular to the tube axis. The frequency shall be varied from 5 to 500 HZ in 15 minutes, then decreased in 15 minutes, for a total time of 30 minutes per cycle. The acceleration shall be held to 2.5G's maximum throughout the frequency cycle. The laser tube shall be operated throughout the vibration test and the power output monitored, and then meet all of the requirements of 3.5.1 and 3.7.1 to 3.7.6.

3.6.4 Shock. - The tube assembly shall be mounted as shown in Figure 2, and shock three times in 4 directions (perpendicular and parallel to tube axis), tor a total of 12 shock pulses. The shock pulse shall have a waveform of approximately a half sine wave, a peak amplitude of 75G's, and a time duration of 11 milliseconds, and then meet the requirements of 3.5.1 and 3.7.1 to 3.7.6.

3.6.5 Humidity. - The tube assembly shall not be damaged or performance degraded by exposure to 95% relative humidity for a period of 24 hours, and subsequently shall meet the performance requirements of 3.7.1 through 3.7.9.

3.6.6 Altitude storage. - The tube assembly shall not be damaged or performance degraded by exposure to pressure conditions of 50,000 feet, and subsequently shall meet the performance requirements of 3.5.1 and 3.7.1 to 3.7.6.

3.7 Performance.

3.7.1 Starting voltahge. - The starting voltage shall not exceed 8,000 VDC and shall not be damaged or performance degraded by starting voltages up to 12,000 VDC.

3.7.2 Starting time. - The starting time shall not exceed 100 milliseconds measured from application of the starting voltage to when the power output of the laser beam reaches 90% of its steady state value.

3.7.3 Power output. - The power output of the laser beam shall be 1.5 ± .5 milli-watts at 632.8 nano-meters wavelength.

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3.7.4 Sustaining current. - The tube current required to sustain laser action to produce the power output of 3.7.3 shall be 6.0 ± .4 mini-amperes, when the voltage drop across the plasma tube assembly falls within the range of 1.2 to 1.6 KVDC as shown in Figure 1.

3.7.5 Frequency sensitivity. - The plasma tube shall not be damaged or degraded with a 75 volt peak-to-peak ripple at a frequency of 5 to 25 KHz superimposed upon the high voltage dc input for a period of five minutes.

3.7.6 Beam alignment. - The laser light beam axis shall be aligned parallel to the longitudinal axis of the mounting cylinder within two milliradians and shall be coaxial to 0.010 inch to the clamping diameter (see Figure 2).

3.7.7 Beam roundness. - The tube must operate in the TEMoo mode, and the beam is required to be round.

3.7.8 Beam divergence. - The beam divergence shall be 1.2 milliradians maximum at the I/e^{z} power points.

3.7.9 Beam exit diameter. - The beam diameter at the l/e^{2} points shall be 1.4 millimeters maximum at the face of the tube.

3.7.10 Voltage-current characteristics. - The laser tube shall have the voltage-current characteristics specified herein as shown on Figure 1.

3.8 Interchangeability. - The tube assembly shall be capable of insertion into and removal from its intended receiving assembly within the drawing tolerances prescribed.

3.9 Marking. - The laser tube shall be permanently marked as identified in MIL-STD-130. The following information is to be marked on the tube.

Manufacturer's Name
Contract Number
Part Number 11746797
"Tube, Laser, Plasma, Helium Neon"
Serial Number
"NSN 6920-00-393-0268"
"Warranty for 18 months"
"Warranty expires:" Date

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. - Unless otherwise specified in the contract ofr purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection

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requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 General provisions. - The component and subassembly inspection requirements of MIL-F-13926 form a part of the quality assurance provisions of this specification. Definitions of inspection terms shall be as listed in MIL-STD-109.

4.1.2 First article (initial Production) approval. - The requirements for first article approval and the responsibility (Government of contractor) for first article testing shall be as specified in the contract. The sample for first article approval tests shall consist of four tube assemblies manufactured in the same manner using the same materials, equipment, processes, and procedures as used in regular production. All parts and materials, including packaging and packing, shall be obtained from the same source of supply as used in regular production. The assemblies shall meet the requirements and tests of Tables I, II, III. Prior to submission for first article test the sample will have successfully passed the requirements of 3.4.1 through 3.4.2 inclusive (un-encapsulated tube).

4.1.2.1 Government testing. - When the Government is responsible for conducting first article approval tests, the contractor, prior to submitting the sample to the Government, shall inspect the sample to insure that it conforms to all the requirements of the contract and submit a record of this inspection with the sample, including certificates of conformance for materials.

4.1.2.2 Contractor testing. - When the contractor is responsible for conducting first article approval tests, the sample shall be inspected by the contractor for all requirements of the contract. The The sample and a record of this inspection, including certificates of conformance for materials, shall be submitted to the government for approval. The Government reserves the right to witness the contractor's inspection.

4.2 Inspection Provision.

4.2.1 Submission of product. - Unless otherwise specified by the contracting officer, inspection lot size, lot formation and presentation of lots shall be in accordance with "Submission of Product" provisions of MIL-STD-105.

4,2.2 Examination and tests.

4.2.2.1 Components and subassemblies. - All components and subassemblies shall be inspected in accordance with the provisions set forth in this specification. Tube, ballast, resistor. connectors prior to encapsulation and the encapsulated assembly shall be inspected as identified in this specification.

4.2.2.2 Fabrication.

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4.2.2.2.1 Certification. - Certification shall be submitted in conformance with 3.1 as part of the first article data stating that the specified fabrication requirements listed below have been met. Certification shall be provided prior to performing final acceptance inspection and remain in effect throughout the contract for Government acceptance. This does not relieve the contractor from inspection throughout the contract production, including recording data and results therefrom. Recorded data and results shall be made available to the Government upon request. A new certification will be required should any changes occur in materials, manufacturing processes and techniques previously certified.

CHARACTERISTIC	REQUIREMENT	METHOD OF INSPECTION
a. Materials	3.1.1	Certification
b. Encapsulation compound	3.1.2	"
c. Fungus	3.1.3	"
d. Internal Mirrors	3.1.4	"
e. Coaxial Reservoir	3.1.5	"
f. Cold Cathode	3.1.6	"
g. Ballast Resistor	3.1.7	"
h. Connectors	3.1.8	

4.2.2.3 100% Inspection. - The requirements and related tests in Table I shall be subjected to 100 percent inspection.

TABLE I - 100% INSPECTION

NO.	CHARACTERISTIC	REQUIREMENT	TEST PROCEDURE
101.	Leakage	3.4.1	4.5.3.1
102.	Burn-in (unencapsulated)	3.4.2	4.5.3.2
103.	Sealing	3.5.1	4.5.4.1
104.	Burn-in (encapsulated)	3.5.2	4.5.4.2
105.	Starting voltage	3.7.1	4.7.1
106.	Starting time	3.7.2	4.7.2
107.	Power output	3.7.3	4.7.3
108.	Sustaining current	3.7.4	4.7.4
109.	Freq. sensitivity	3.7.5	4.7.5
110.	Beam alignment	3.7.6	4.7.6
111.	Beam roundness	3.7.7	4.7.7
112.	Beam divergence	3.7.8	4.7.8
113.	Beam exit diameter	3.7.9	4.7.9
114.	Volt-current curve	3.7.10	4.7.10
115.	Interchangeability	3.8	4.8
116.	Marking	3.9	4.9

4.2.2.4 Environmental. - Three assemblies shall be selected at random from each 50 produced or from each month's production, whichever occurs first, and subjected to the tests in Table II. Each sample shall then meet all the tests specified under Test Methods and Procedures; paragraph 4.5.

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TABLE II

NO.	CHARACTERISTIC	REQUIREMENT	TEST PROCEDURE
201.	Storage temperature	3.6.1	4.6.1
202.	Operating temperature	3.6.2	4.6.1
203.	Vibration	3.6.3	4.6.2
204.	Shock	3.6.4	4.6.3

4.2.3 Special sampling. - Four tube assemblies shall be selected at random and meet the requirements and tests of Table III. These tests shall be conducted as follow-up to the first article samples at 50% completion of the production contract quantity.

TABLE III

No.	CHARACTERISTIC	REQUIREMENT	TEST PROCEDURE
301.	General specification	3.2	Visual and Tactile MIL-F-13926
302.	Humidity	3.6.5	4.6.4
303.	Altitude - storage (50,000 feet)	3.6.6	4.6.5
304.	Reliability	3.3	4.5.2

4.3 Acceptance and rejection. - Rejected lots shall be screened for all defective characteristics. Removal or correction of defective units and re-submittance of rejected lots shall be in accordance with "Acceptance and Rejection" as specified in MIL-STD-105.

4.4 Inspection euipmetn. - Except as otherwise provided for by the contract, the contractor shall supply and maintain inspection in accordance with the applicable requirements of MIL-I-45607.

4.4.1 Government-furnished inspection equipment. - Where the contract provides for Government furnished test equipment, supply and maintenance of test equipment shall be in accordance with applicable requirements specified in MIL-I-45607.

4.4.2 Contractor-furnished inspection equipment.

4.4.2.1 Government design. - All inspection equipment specified by drawing number in specifications or SQAP'S forming a part of the contract shall be supplied by the contractor in accordance with technical data included in the Technical Data Package List.

4.4.1.1 Contractor design. - The contractor shall design and supply inspection equipment compatible with the "Test Methods and Procedures" specified in this specification and with the component inspection procedures specified in "Examination" and "Test Facilities" requirements of MIL-F-13926. Since tolerance of test equipment is normally considered to

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be within 1077 of the product tolerance for which it is intended, this inherent error in the test equipment design must be considered as part of the prescribed product tolerance limit. Thus, concept, construction, materials, dimensions and tolerances used in the design of test equipment, shall be so selected and controlled as to insure that the test equipment will reliably indicate acceptability of a product which does not exceed 90% of the prescribed tolerance limit, and permit positive rejection when non-conforming. Construction shall be such as to facilitate routine calibration of test equipmetn. The test equipment selected by the contractor to be utilized to perform the tests of 4.7, shall be submitted to the Government for approval with appropriate sketches and accuracies of measured parameters prior to first article tests. The proposed test equipment documentation shall also include all set-up and operating procedures used by the contractor to test the requirements of this specification.

4.5 Test methods and procedures.

4.5.1 Fabrication.

4.5.1.1 Certification.- 4.5.1.1 Certification. - Inspection records submitted in compliance with the materials requirements in 3.1 for final acceptance shall show evidence that the tube has conformed to the requirements of 3.1.1 through 3.1.8 and 4.2.2.2.1.

4.5.2 Reliability. - The reliability of the plasma tube assembly shall be demonstrated in conjunction with First Article and Special Sampling (4.2.3) testing to determine compliance with 3.3. The pre and post encapsulation tests of 4.5.3 and 4.5.4 shall have been satisfactorily accomplished prior to the reliability tests. The tuve shall be operated for five cycles in continuous mode (50 minutes on - 10 minutes off constitutes one cycle) after exposure to each of the non-operating environments and during exposure to the operating environments in addition to those performance tests specified in 4.6. After completion of all required environmental test the samples shall be operated in the continuous mode at room ambient conditions in order to accumulate a minimum of 1000 hours of operation on each sample tube. The sample tubes shall be tested until the total hours accumulated meet the criteria of Table IV. During this room ambient temperature test the power of the laser beam and the starting time shall be measured (all samples) after each 1000 total accumulated hours (recorded test results). After successfully comply with Table IV the tube assemblies shall meet the requirements of 3.7.

TABLE IV

TOTAL HOURS ACCUMULATED	RELIABILITY ACCEPT #	FAILURES REJECT #
6,020	0	4
12,195	1	4
18,077	2	4
23,810	3	4

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4.5.2.1 Data collection. - Data will consist of all malfunctions, number of hours completed when malfunction occurred, nature and cause of malfunction, time required to correct malfunction, corrective action required, item serial number and environmental exposure.

4.5.2.2 Failure definition. - A failure is defined as any malfunction which results in an inoperative tube or a tube which is operative outside the limits of this specification. In determination of chargeable reliability failures, the following clarification shall apply.

4.5.2.3 Assignable cause. - If upon failure analysis (to determine the cause of malfunction) an-assignable cause (related to manufacturing process, procedure, etc.) is determined as the basis of the malfunction, the malfunction will not be chargeable as a reliability failure; however, the correction to the manufacturing process or procedure (et al) shall be accomplished.

4.5.2.4 Unassignable cause. - If upon failure analysis no assignable cause is evident and the malfunction is considered random in nature, the malfunction shall be chargeable reliability failure and as such must be considered in the above demonstration. The dexisions as to chargeable reliability failure must be coordinated with the procuring activity, Frankford Arsenal Reliability & Maintainability Engineering Branch (SARFA-QAF-R).

4.5.3 Pre-encapsulation.

4.5.3.1 Leakage. - To determine compliance with 3.4.1, the tube shall be held in a suitable container and immersed completely in water (H_20) so that both ends of the tubes are at approximately the same depth. The water shall then be heated by any suitable means to being the water to a boil (approximately 212°F). The water shall be boiled-for four hours at which time the tubes shall be removed and allowed to dry and cool. The tubes shall be visually inspected for damage. Subsequently the tube shall be subjected to the tests of 4.5.3.2.

4.5.3.2 Burn-in (unencapsulated). -To determine compliance with 3.4.2, utilize a power supply simulating the Power Supply Module, F11739861 with Ballast Resistor (see 3.1.7) connected in series with the tube. Operate the laser tube through 24 cycles of 50 minutes on and 10 minutes off. Subsequently the tube shall be subjected to the tests of 4.7.1 through 4.7.5 inclusive to determine compliance with the requirements of 3.7.1 through 3.7.5.

4.5.4 Post encapsulation tests.

4.5.4.1 Sealing. - A test fixture containing an adapter type box to encapsulate and pressurize the entire tube area extending from the flanged portion of the clamping area (see Figure 2) to the output end shall be provided to determine compliance with 3.5.1. With the tube assembly secured in the pressure fixture containing a pressure gage accurate to within 10% of leakage rate of 0.2 psig per two hours, the box shall pressurized at 5 psig for a period of two hours to determine compliance with 3.5.1.

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4.5.4.2 Burn-in encapsulated. - To determine compliance with 3.5.2, utilize a power supply simulating the Power Supply Module F11739861. Operate the tube assembly through 24 cycles of 50 minutes on and 10 minutes off. Subsequent to the burn-in test the tube assembly shall be subjected to all of the tests in 4.7 to determine compliance with the requirements of 3.7, 3.8 and 3.9.

4.6 Environmental tests. - To determine compliance with all of the requirements of 3.6 the tube assembly shall be subjected to the tests specified in 4.6.1 through 4.6.5. There shall be no evidence of corona when the laser tube is subjected to each of the specified environmental test. Data shall be recorded prior to operation, during operation and post operation in accordance with all of the environment test procedures specified herein. Available post-test data may be used as pretest data for the succeeding test. Complete environmental test data shall be recorded and submitted as a part of reliability data.

4.6.1 Storage and operating temperature. - To determine compliance with 3.6.1 and 3.6.2 this test shall be performed in accordance with MIL-STD-81O, Method 501 and 502, Procedure I respectively. Prior to this test the tube assemblies shall have successfully passed the performance tests of 4.7.1 through 4.7.4. The tube assemblies shall be place in the temperature chamber. The chamber temperature shall be lowered to the storage temperature or -65°F and remain at this extreme temperature for at least 24 hours. The chamber temperature shall then be raised to the operating temperature of 0°F and remain at this temperature for at least four hours. While the tube assemblies are 0°F they shall be subjected to performance tests of 4.7.1 through 4.7.4. After the tube assemblies have successfully passed the performance tests, they shall be energized and remain on throughout the period that the chamber is raised from 0°F to 100°F. During the period they are energized, power output of the laser beam shall be monitored to determine compliance with the requirement of 3.7.3. When the chamber temperature reaches the storage temperature of +155°F, the chamber shall remain at this temperature for not less than 24 hours. The chamber temperature shall then be lowered to the operating temperature of 100°F and remain at this temperature for not less than four hours. The tube assemblies shall again be subjected to the operational test 4.7.1 through 4.7.4. After being subjected to these performance tests the chamber shall be returned to room ambient temperature while the tube assemblies are energized and the output power shall be monitored. At room ambient temperature the tube assemblies shall be subjected to all of the performance tests of 4.7 to determine compliance with the requirements of 3.7.

4.6.2 Vibration. - To determine compliance with 3.6.3, the test shall be performed in accordance with MIL-STD-810, Method 514.1, Procedure VIII, Figure 514.1-6 Curve "Y" and the conditions specified in 3.6.3. During vibration, the tube assemblies shall be energized and the power output monitored. Subsequently, the tube shall be subjected to the performance test of 4.5.4.1 and 4.7.1 through 4.7.6, inclusive to determine compliance with the requirements of 3.5.1 and 3.7.1 through 3.7.6.

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4.6.3 Shock. - To determine compliance with 3.6.4, the test shall be performed accordance with MIL-STD-810, Method 516.1, Procedure 1, Figure 516.1-2, at 75 G's for 11 milliseconds. After the shock tests the tube assemblies shall be subjected to the performance tests of 4.5.4.1 and 4.7.1 through 4.7.6 inclusive to determine compliance with the requirements of 3.5.1 and 3.7.1 through 3.7.6.

4.6.4 Humidity. - To determine compliance with 3.6.5, the test shall be performed in accordance with MIL-STD-810, Method 507, Procedure I, with the following excption. The test duration. shall be a minumum of 24 hours at 95% relative humidity and limited to one cycle. Subsequently the tube assemblies shall be subjected to performance tests of 4.7.1 through 4.7.9 inclusive to determine compliance with the requirements of 3.7.1 through 3.7.9.

4.6.5 Altitude.- 4.6.5 Altitude. - To determine compliance with 3.6.6, the test shall be performed in accordance with MIL-STD-810, Method 500, Procedure 1. Subsequent to this test the tube assemblies shall be subjected to the performance tests of 4.7.1 through 4.7.6 inclusive to determine compliance with the requirements of 3.7.1 through 3.7.6.

4.7 Performance characteristics. - Compliance with 3.7 shall be determined by means of tests specified in 4.7.1 through 4.7.10. The tests shall be performed utilizing a power (high voltage) supply simulating that shown on Drawing F11739861. The high voltage power supply shall be capable of varying the output voltage of between 5 KVDC to 12 KVDC to an accuracy or +100 VDC unloaded. Means shall be provided to dump the full voltage across the laser tube assembly. The voltage drop across tube assembly during lasing shall be measured to an accuracy of ±50 VDC. The current shall be measured to an accuracy of ±0.1 milliamps D.C. An oscilloscope shall be connected such that the time between the applied voltage and the resulting laser output can be measured. A He-Ne power meter capable of measuring 2 mini-watts and having an external commector for an oscilloscope shall be provided to measure the waveform of the laser pulse as seen by the detector. The power meter shall be capable of measuring 638.8 Nanometers laser power to ± 0.10 mini-watt accuracy. All test equipment shall be calibrated and traceable to the National Bureau of Standards.

NOTE: If the laser tube fails to lase or has insufficient power output allow a maximum of 30 minutes warm up time before performing 4.7.1 thru 4.7.10.

4.7.1 Starting voltage. - Utilizing the test equipment,specified in 4.7, apply 7.5 KVDC across tube assembly. View the power wave form on the scope. The output power shall meet the requirement of 3.7.3 within the time specified in 3.7.2. If tube does not lase with 7.5 KVDC applied, increase the voltage to 8 KVDC.Measure power output and starting time to determine compliance with the requirement of 3.7.1. subsequent to Successfully passing the previous test, the voltage shall be increased to 12 KVDC

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and applied to the tube assembly. The plasma tube shall lase and meet the requirements of 3.7.2 and 3.7.3. This shall be repeated 10 times with at least a 30 second rest between applying these over-voltage pulses.

4.7.2 Starting time. - This test shall be performed in conjunction with 4.7.1 to determine compliance with the requirements of 3.7.2.

4.7.3 Power output. - This test shall also be performed in conjunction with 4.7.1 to determine compliance with the requirements of 3.7.3. A calibrated He-Ne power meter shall be used (with an output for an oscilloscope) to monitor the power to within 0.10 mini-watts.

4.7.4 Sustaining current. - With the voltage developed across the tube assembly of between 1.2 KVDC to 1.6 KVDC (see Figure 1) and the power output within the requirements of 3.7.3. The current through the tube shall be measured to determine compliance with the requirements of 3.7.4.

4.7.5 Frequency sensitivity. - With a 75 volt peak-to-peak ripple voltage at a frequency varying between 5 to 25 KHZ superimposed on the input for a period of not less than five minutes, the tube assembly shall be subsequently subjected to the test of 4.7.1 to 4.7.4 inclusive to determine compliance with 3.7.5.

4.7.6 Beam alignment. - Beam alignment shall be accomplished by establishing a true line of sight from the geometric center of the clamping diameter to a wall target. With the tube assembly held as shown in Figure 2, rotate the tube assembly 360° and note diameter of the circle transctibed by thelaser spot. The diameter shall not exceed that specified in 3.7.6 (2mini-radians). - The laser beam existing from the tube assembly shall be centered within 0.01 inch of the center of the clamping diameter.

4.7.7 Beam roundness. - Beam roundness is required to verify TEMoo mode. (Fundamental mode). Beam roundness is accomplished by visually observing the projected beam using a microscope objective with at least a 1 cm aperture and a short focal lengtth (approximately .25cm) to enlarge the beam spot. The observed spot shall be round in shape without discontinuities in the intensity across the diameter.

4.7.8 Beam divergence. - Beam divergence shall be calculated using the maximum diameter as determined in 4.7,7 and the distance between the laser tube and the wall target in the following relationship:

a. Divergence rad = d ÷ R, where:

R = distance from laser tube to wall target.
d = maximum diameter of laser spot.

The results shall not exceed the mill-radians specified in 3.7.8.

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4.7.9 Beam exit diameter. - This test shall be performed utilizing a fixed aperture (1.4mm) at the exit beam and measuring the power through the aperture to determine compliance with the requirements of 3.7.9.

4.7.10 Voltage current characteristics. - This curve shall be established at room ambient temperature, with the output power maximized and a voltage drop across the tube assembly of between 1.2 and 1.6 KVDC. Vary the tube current from five to 10 mini-amps and record voltage drop changes. The curve shall be as specified in Figure 1.

4.8 Interchangeability. - This test shall be performed utilizing the interchangeability gages conforming to Drawings B11739401 and B11739402 to determine compliance with 3.8.

4.9 Marking. - To determine compliance with 3.9, a visual examination shall be conducted as specified in MIL-STD-130.

5. PREPARATION FOR DELIVERY

5.1 Preparation for delivery shall be in accordance with the requirements of MIL-E-75, package group 23. For level A preservation-packaging, each laser tube shall be inclosed in a sealed water-vapor-proof barrier bag filled with desiccant.

6. NOTES

6.1 Intended use. - The laser tube assembly is used as part of a Laser Tank Gunnery Trainer by the US Army. The Laser Tank Gunner Trainer is a training device which will mount in the M73 Machine Gun Mount of the M48A3 Tank, M60 Tank, M60A1 Tank and AR/AW, M551 and permit year around training indoors or outdoors in the techniques of zeroing, accuracy and rapidity of laying the gun using direct fire sights.

6.2 Ordering data. - Procurement documents should specify the follow(-ing:

(a) Title, number and date of this specification.

(b) Applicable stock number.

(c) Requirements. for first article testing (see 4.1.2).

6.3 Warranty. - The laser tube manufacturer shall warrant the tube for at least 18 months from the date of shipment. This warranty applies where the tube is operated within the conditions specified herein.

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6.4 Definitions.

6.4.1 1/e^2 Power points. - A l/e^2 power point is that point where the diameter of the beam encloses I/e^2 or 0.865 of the power of the total beam.

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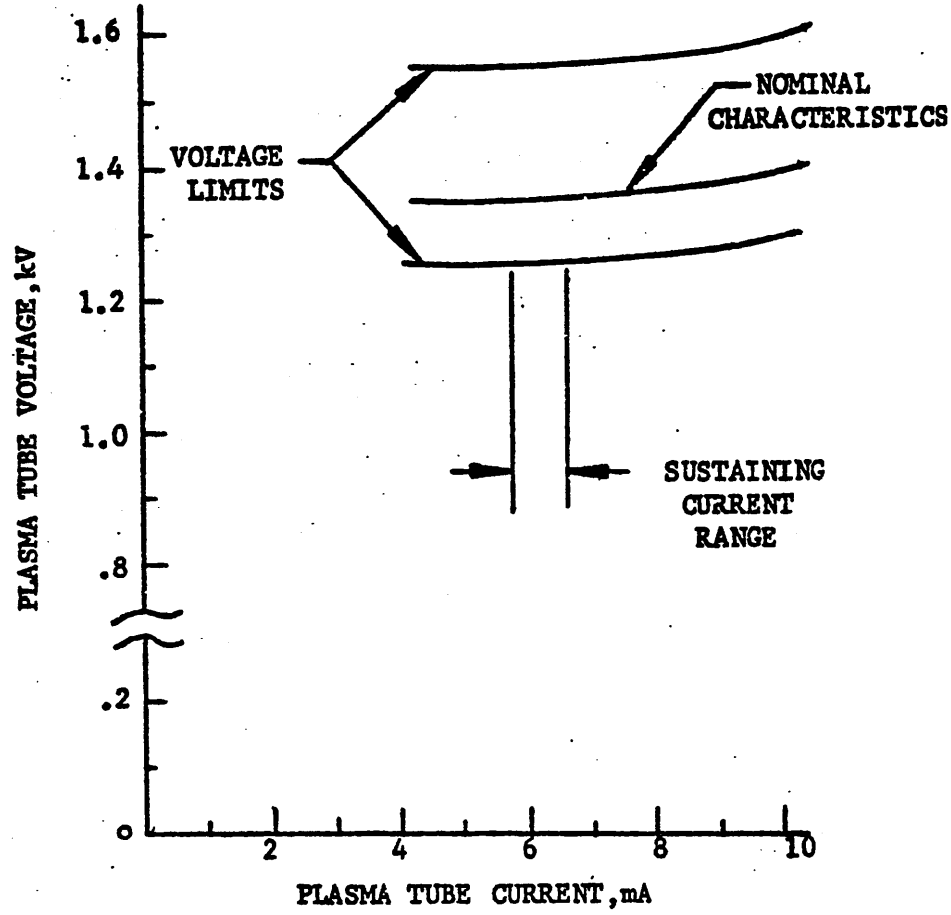


FIGURE 1 Voltage-current characteristics

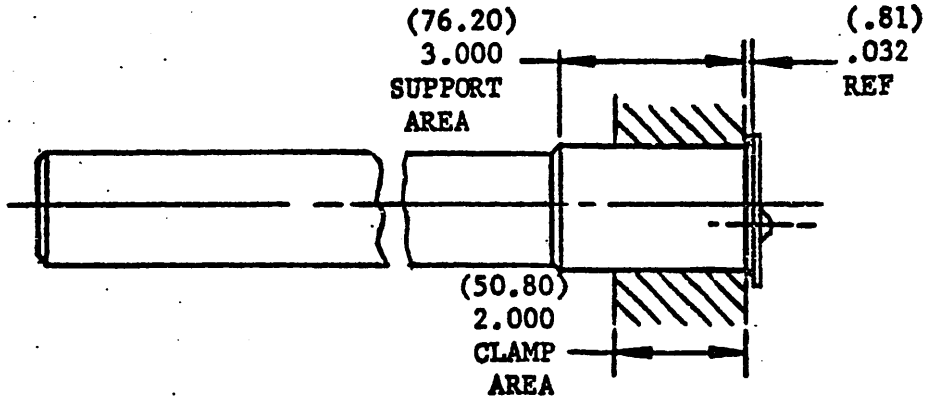


FIGURE 2 MOUNTING CONFIGURATION

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

OMB Approval
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DOCUMENT IDENTIFIER AND TITLE
MIL-T-48604(MU) Tube, Laser, Plasma, Helium Neon: 11746797

NAME OF ORGANIZATION AND ADDRESS

CONTRACT NUMBER

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE DOCUMENT CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?
   A. GIVE PARAGRAPH NUMBER AND WORDING.

   B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY DOCUMENT REQUIREMENT CONSIDERED TOO RIGID

3. IS THE DOCUMENT RESTRICTIVE?

   ☐ YES ☐ NO *(If "Yes", in what way?)*

4. REMARKS

SUBMITTED BY *(Printed or typed name and address - Optional)*

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